Kinematic Safety Analysis of Deep Reinforcement Learning Policies for Robotic Polishing

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***Abstract*—Robotic surface polishing requires sweeping a tool across a free-form workpiece while maintaining contact, along a path whose spacing governs how uniformly material is removed. We present a deep reinforcement learning framework in which the coverage path is generated as isolines of a geodesic distance field on the workpiece mesh and executed under contact by a 5-DOF operational-space policy on a 6-DOF UR20.**

***Index Terms*—Deep Reinforcement Learning, Robotic Manipulators, Coverage Path Planning, Robotic Polishing**

I. Introduction

Surface polishing operations on complex workpieces are, in the majority, still performed by human operators, because they demand a degree of online adaptation to the geometry of the part that offline-coded trajectories do not provide [1]. Automating them means sweeping a tool across a free-form surface while maintaining contact, along a path whose *spacing* governs how uniformly material is removed. Coverage quality is therefore decided twice: once when the path is planned, and again when a controller tries to follow it while staying on the surface.

We therefore present a deep reinforcement learning framework in which the coverage path is generated in the intrinsic surface metric and executed under contact by a 5 Degrees of Freedom (DOF) operational space policy on a 6-DOF UR20. Our contributions are as follows.

- A learned polishing framework whose coverage plan is intrinsic to the surface and whose execution is closed on contact.
- An *area-based* coverage metric reported against the *path ceiling*, the coverage attainable under perfect tracking. Together they separate the quality of the plan from that of the policy, and expose the gap that waypoint completion hides.
- An analysis of kinematic safety under prescribed coverage, given a prescribed placement of the workpiece, comparing a joint-velocity effort reward with a conditioning-proximity reward, and showing that the ill-conditioned region is fixed by the placement together with the initial arm posture.

II. Related Work

A. Coverage Path Planning on Free-Form Surfaces

In robotic polishing, trajectory spacing directly dictates the consistency of material removal. Gong, Li, Zhang, Fan, and Li [2] note that traditional trajectory generation approaches such as iso-parametric and iso-planar methods often fail to account for local surface curvature, resulting in uneven step-over distances and degraded polishing quality. Within general Coverage Path Planning (CPP), generating paths via 2D-to-3D projections or parallel plane intersections can result in geometric distortion on high-curvature geometries [3]. Furthermore, relying on extrinsic Cartesian reference frames may produce uneven surface discretization [4]. To overcome these errors, recent literature advocates for intrinsic parameterizations such as replacing Euclidean metrics with arc-length uv spaces [4] or exact geodesic distances [3]. This geometric awareness is equally critical for learned motion policies; assuming a flat planning space and relying solely on low-level control to maintain surface contact fails to exploit the intrinsic geometry of the workpiece, thus limiting generalization [1].

B. Kinematic Safety in Learned Manipulation

Proximity to singular configurations degrades dexterity and drives joint velocities to grow without bound [5]. In learned manipulation, safety is typically enforced through either structural constraints or reward penalties. Structural approaches guarantee safety by construction, remapping learned actions onto safe velocity intervals via trajectory generators [6], or constraining the action space to a safe manifold [7]. Under such hard constraints, kinematic safety becomes an invariant property of the controller rather than a learned behavior. On the other hand, reward-based methods lead manipulators away

from ill-conditioned postures by penalizing standard proximity measures, such as Yoshikawa's manipulability [8].

III. Methodology

A. Surface Representation and Coverage Path

Free-form workpieces present a fundamental challenge for coverage planning: a constant step-over defined in a Cartesian projection does not correspond to a constant spacing on the surface itself. This discrepancy becomes particularly pronounced in regions of high curvature, leading to non-uniform polishing overlap. To overcome this limitation, the proposed method generates the polishing trajectory directly from the intrinsic geometry of the workpiece by means of a geodesic distance field, ensuring that the prescribed spacing is measured along the surface rather than in Euclidean space.

The polishable region is obtained by extracting the region of interest from the mesh. Faces whose outward normal is oriented toward the tool are retained, producing the accessible polishing patch. Because geodesic distance computation requires a connected manifold, only the largest connected component is preserved.

A geodesic distance field, denoted by $\phi(v)$ is then computed over the polishing patch using the Heat Method [9].The computation starts from a seed point placed at one end of the workpiece, whose location determines the overall direction of the polishing bands. The trajectory is generated by extracting equally spaced isolines of the distance field, yielding a constant step-over measured directly on the surface (Fig. 1).

Each isoline is ordered and uniformly resampled according to its arc length to produce evenly spaced waypoints. Consecutive bands are then connected through the shortest transition while alternating their traversal direction, resulting in a continuous serpentine trajectory covering the entire polishing region.

Every waypoint stores its Cartesian position together with the corresponding surface normal, obtained by nearest-neighbor query on the mesh. During execution, these quantities are used by the controller to construct the task frame that guides the polishing motion.

B. Markov Decision Process Formulation

The surface polishing task is modeled as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma)$. At each timestep t, the agent receives an observation $s_t \in \mathcal{S}$, executes an action $a_t \in \mathcal{A}$, receives a scalar reward r_{total}, and transitions to a new state s_{t+1} according to the environment dynamics $\mathcal{P}$.

1) Observation Space: The observation exposes task-relevant information independently of the joint configuration. It consists of the robot state (end-effector position, quaternion orientation, linear and angular velocities) as well as the task state. This task state includes the vector from the tool center point to the active waypoint, the signed distance to the workpiece surface and the local surface normal. Additionally, the observation contains the previous action and the reciprocal condition number w of the task Jacobian, per Eq. (2).

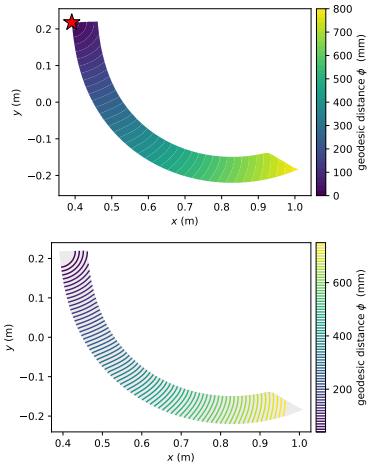


Fig. 1. Planning the coverage bands in the surface. **Top:** the geodesic distance field ϕ, computed over the working region by the Heat Method from a seed at one end of the workpiece. **Bottom:** isolines of ϕ extracted at 10 mm intervals. Because ϕ is a distance field, consecutive isolines are separated by that interval *as measured on the surface* rather than in a projection.

2) Action Space and Control Strategy: Rather than commanding joint torques or velocities directly, the agent outputs normalized velocity commands expressed in a local surface-referenced frame. The action space is defined as $a_t \in [-1, 1]^4$.

The action separates into two components. The first three elements specify the desired Cartesian velocity in a frame whose axes are the heading toward the active waypoint projected onto the tangent plane, the transverse tangent direction, and the surface normal. The fourth component commands the available self-motion associated with the one-dimensional null space arising from the task redundancy.

The polishing tool is axisymmetric, so its spin about the longitudinal axis is irrelevant to the process. A manipulator is redundant with respect to such a task when it has more than 5 joints [10]. Our 6-DOF UR20 therefore possesses one redundant degree of freedom of task origin rather than intrinsic to the arm. The commanded velocity is mapped to joint velocities through a damped least-squares inverse kinematics on the resulting 5×6 task Jacobian [11],

$$\dot{\mathbf{q}} \;=\; \mathbf{J}^{\top}\left(\mathbf{J}\mathbf{J}^{\top} + \lambda^{2}\mathbf{I}\right)^{-1}\dot{\mathbf{x}} \;+\; \beta\, a_t^{(4)}\, \hat{\mathbf{n}}(\mathbf{q}), \tag{1}$$

where $\mathbf{J} \in \mathbb{R}^{5\times 6}$ is the task Jacobian, $\dot{\mathbf{x}} \in \mathbb{R}^5$ is the commanded task velocity stacking the Cartesian velocity requested by the first three action components with a proportional correction that aligns the tool axis to the surface normal, λ is the damping factor, β the self-motion scale, and $\hat{\mathbf{n}}_{\mathbf{6\times 1}}(\mathbf{q})$ the unit vector spanning the null space of $\mathbf{J}_{\mathbf{5\times 6}}$, which is one-dimensional away from singularities. Since $\mathbf{J}\hat{\mathbf{n}} = \mathbf{0}$, motion along $\hat{\mathbf{n}}$ does not displace the task target. Because the tool is mounted coaxially with the last joint, the tool centre point lies on that joint's axis. Consequently, the last column of $\mathbf{J}$ vanishes and no other column depends on q_6. Hence, $\hat{\mathbf{n}} = \mathbf{e}_6$, meaning that self-motion only spins the tool without changing

$\mathbf{J}$ or its conditioning. Finally, the resulting joint velocities are saturated at the robot velocity limits before execution.

Lastly, we define the maximum and minimum singular values of $\mathbf{J}$ as $\sigma_{max} = \sigma_1$ and $\sigma_{min} = \sigma_5$. They are the gains of the map from joint velocity to task velocity along its principal directions. Specifically, σ_{min} is the gain along the direction in which the arm is *least* able to move, and it vanishes when $\mathbf{J}$ loses rank. Therefore, the conditioning measure exposed in the observation is their ratio,

$$w \;=\; \frac{1}{\kappa(\mathbf{J})} \;=\; \frac{\sigma_{min}}{\sigma_{max}} \;\in\; [0,1], \tag{2}$$

C. Reward Function Design and Kinematic Safety

The reward is a linear combination of task incentives, safety penalties and regularization terms. The total reward is

$$r_t = r_{task} + r_{align} + r_{safe} + r_{smooth} + r_{off} \tag{3}$$

with a terminal penalty on episode failure. Weights are in Table I.

1) Task and Coverage Rewards: The primary objective is to drive the end-effector along the geodesic waypoints while maintaining constant surface contact. The task reward is formulated as:

$$r_{task} = r_{prog} + r_{wp} + r_{cont} + r_{cov}$$

Here, $r_{prog} = k_p(d_{t-1} - d_t)$ provides a dense gradient proportional to the distance reduced towards the active waypoint, while r_{wp} is a sparse bonus granted upon reaching it. Surface contact is encouraged through $r_{cont} = k_c\, g_m$, where $g_m \in [0,1]$ only pays for new forward progress. Specifically, the system only rewards the reduction of the distance to the active waypoint, not rewarding policy actions that halts or oscillates back and forth. Finally, $r_{cov} = k_{cov}\,\Delta A_t$ rewards the area newly grinded by the tool while it is in contact.

2) Tool Alignment and Kinematic Redundancy: Polishing is inherently a 5-DOF task, as rotations about the tool's longitudinal axis do not affect the process. Accordingly, the proposed formulation regulates only the task-relevant degrees of freedom through the alignment penalty

$$r_{align} = -k_n(1 - \hat{t} \cdot (-\hat{n}))$$

This term aligns the tool axis ($\hat{t}$) with the inward surface normal ($-\hat{n}$). By tracking the normal, the task becomes 5-DOF, leaving the rotation about the tool axis as the redundant degree of freedom, which here is the last joint of the robot (Eq. (1)).

3) Kinematic Safety and Action Regularization: To operate safely, the policy must avoid ill-conditioned configurations near kinematic singularities. A common choice is to penalize proximity to singularities through a manipulability measure [5], [8]. In our framework the agent interacts with the robot through a damped least-squares differential inverse kinematics controller whose output is the joint velocity. Consequently, we express the safety objective in terms of the normalized joint-velocity effort:

$$r_{safe} = -k_{\dot{q}}\,\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\dot{q}_i}{\dot{q}_{max}}\right)^2 \tag{4}$$

where N is the number of robot joints, $\dot{q}_i$ is the commanded velocity of joint i, $\dot{q}_{max,i}$ is a common velocity scale. Because ill-conditioned configurations require progressively larger joint velocities to maintain the desired end-effector motion, minimizing this term encourages the policy to drive away from singular regions. Since the redundant degree of freedom leaves the conditioning unchanged, the policy can lower this effort only by slowing down or by deviating laterally from the path.

The damped least-squares inverse of Eq. (1) attenuates the component of the commanded velocity along the i-th principal direction by $\sigma_i^2/(\sigma_i^2 + \lambda^2)$, so the damping factor supplies a physical scale: at $\sigma_{min} = \lambda$ half of the commanded motion along the weakest direction is lost, and at $\sigma_{min} = 3\lambda$ one tenth of it. We therefore call a configuration *ill-conditioned* when $\sigma_{min} < 3\lambda$, and we use no other criterion. The rows of $\mathbf{J}$ are unweighted, with units of $\mathrm{m\,rad^{-1}}$ for translation and $\mathrm{rad\,rad^{-1}}$ for rotation. Since λ acts on this same matrix, 3λ is the controller's own threshold. Rescaling the rotational rows by $0.3\,\mathrm{m}$ or $0.12\,\mathrm{m}$ preserves the ordering of the 566 waypoints by σ_{min} (Spearman $\rho \geq 0.99$) and retains at least $95\,\%$ of the ill-conditioned waypoints among those ranked worst. For the safety analysis we also implement a conditioning-proximity penalty in the manner of the reward-based mechanisms of [5], [8]:

$$r_{sing} = -k_w \max\!\big(0,\; 3\lambda - \sigma_{min}\big)^2. \tag{5}$$

Three regularization terms complete the reward. A smoothness penalty $r_{smooth} = -k_s\,\|a_t - a_{t-1}\|^2$ discourages abrupt action changes as a proxy for jerk. A non-gated off-surface penalty $r_{off} = -k_o\,\max\!\big(0,\,|d_{surf}|\big)$ grows linearly once the tool is not in contact. Because it is not motion-gated, it fires even when the tool is stationary, preventing the agent from escaping the surface by hovering. Finally, a terminal penalty of -500 is applied if the tool crosses a catastrophic distance from the surface, ending the episode.

D. Reinforcement Learning Architecture

The agent is trained with Proximal Policy Optimization (PPO) [12], using the implementation of Stable-Baselines3 [13]. Policy and value functions are separate Multilayer Perceptrons (MLPs) with two hidden layers of 64 units that utilize the hyperbolic tangent activation function, fed by a feature extractor that concatenates the dictionary observations. To stabilize learning across diverse physical units and ensure bounded inputs, observations and rewards are normalized by running estimates of their mean and variance, which are accumulated over all training samples and clipped. Crucially, to prevent distribution shifts during deployment, these running statistics are updated exclusively during the training phase and strictly frozen during evaluation. The checkpoint achieving the largest polished area during a deterministic evaluation episode is retained, and this episode is reported.

IV. Experimental Setup

The manipulator is a 6-DOF UR20 carrying a polishing cylindrical tool whose contact disc has a diameter of 30 mm. The workpiece is a curved C-shaped shell segment mesh with an area of 618×440 mm and 21 mm of height.

With a 10 mm of both step-over and waypoint spacing, the path comprises 71 bands and 566 waypoints over 5.709 m (Fig. 2). A waypoint is reached when the tool comes within 8 mm of it, strictly less than the waypoint spacing, to reduce the possibility of chaining waypoints without motion. The simulation runs in PyBullet at 240 Hz and the policy acts every ten physics steps (24 Hz), so at maximum Cartesian speed the tool advances 2.08 mm per control step. An episode lasts at most 4500 control steps, corresponding to 187.5 s.

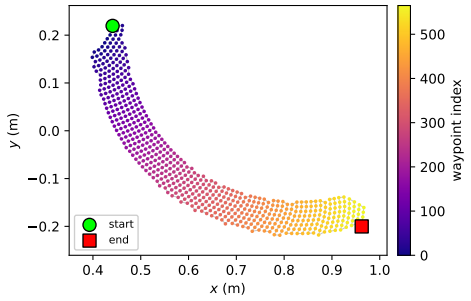


Fig. 2. The resulting toolpath. Colour encodes traversal order.

The kinematic difficulty of the task is set by where the workpiece is located and by the arm posture in which the task starts (Fig. 3). We evaluate at a nominal placement ($x_{wp} = 0.85$ m), where the path is comfortably trackable, and at a stressed one ($x_{wp} = 0.70$ m), where a fraction of the waypoints cannot be traversed at nominal speed from the posture set by the environment reset, the same in every run. For the safety ablation, we report the saturation rate, defined as the fraction of control steps in which a joint reaches its velocity limit within the ill-conditioned region. Each condition is trained for 2×10^6 steps across eight parallel environments with five different seeds to measure training variability. Results are mean $\pm$ s.d. over seeds, compared with the two-sided Mann–Whitney U test ($n = 5$ per arm); each run takes about 30 min on an 8-core CPU. Remaining hyperparameters are in Table I.

To position the learned policy performance before the existing approaches, the same geodesic path is executed by a classical, learning-free pipeline. We use coverage path planning followed by inverse kinematics. For each of the 566 waypoints, the target pose places the tool at a fixed distance along the local surface normal, with the tool axis aligned to it. We obtain the joint configuration using position-level IK seeded from the previous waypoint's solution, ensuring a deterministic trajectory along a single inverse-kinematics branch. The resulting configurations are executed under position control, subject to the same joint-velocity limit ($0.5\,\mathrm{rad\,s^{-1}}$) and simulation environment as the agent. Both methods start from the same initial posture and operate under the same time constraints.

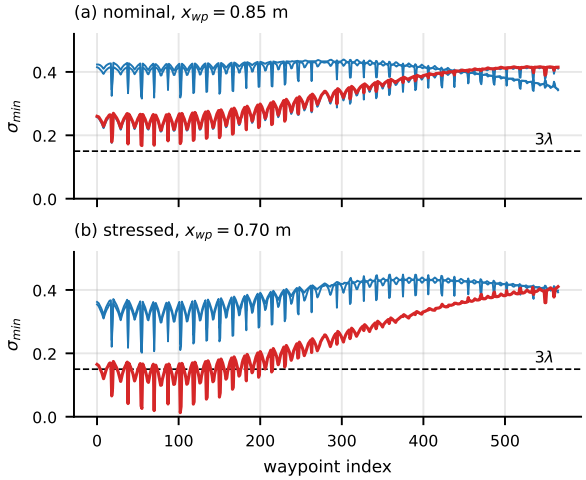


Fig. 3. Smallest singular value σ_{min} of the task Jacobian along the path for each inverse kinematics solution of the arm that completes it without switching posture. Red: posture set by the environment reset, used in all experiments; blue: other postures within joint limits and free of collision. Dashed: 3λ. **(a)** Nominal placement. **(b)** Stressed placement.

We tested distances along the normal from 0 to 12 mm and selected the best performing value, resulting in 0 mm for the nominal condition and 2 mm for the stressed condition.

TABLE I
Controller, PPO, and Reward Parameters

Parameter	**Value**
Cartesian velocity scale v_{max}	$0.05\,\mathrm{m\,s^{-1}}$
Joint velocity limit $\dot{q}_{max}$	$0.5\,\mathrm{rad\,s^{-1}}$
DLS damping λ / orient. gain	0.05 / 3.0
Self-motion scale β	$0.1\,\mathrm{rad\,s^{-1}}$
Timesteps / parallel envs	2×10^6 / 8
Learning rate	1×10^{-4}
Rollout / minibatch / epochs	512 / 128 / 5
Discount γ / GAE λ	0.99 / 0.95
Entropy coef. / clip range	0.005 / 0.1
Reward term	**Weight**
Coverage r_{cov}	$k_{cov} = 2000$
Contact r_{cont}	$k_c = 2$
Progress r_{prog}	$k_p = 200$
Waypoint bonus r_{wp}	1
Alignment r_{align}	$k_n = 1$
Joint-velocity effort r_{safe}	$k_{\dot{q}} = 8$
Ill-conditioning r_{sing}	$k_w = 30$; $3\lambda = 0.15$
Smoothness r_{smooth}	$k_s = 0.3$
Off-surface r_{off}	$k_o = 50$
Terminal	-500

V. Results and Discussion

A. Coverage

By design, the geodesic-based waypoint generation leaves a minor portion of the surface uncovered, creating a theoretical coverage limit. Assuming perfect tracking, simulating the disc grinding along the waypoints and weighting faces by area, the path covers 97.6 % of the working region.

Coverage is only as credible as its contact criterion: when the physics engine first reports disc–workpiece contact, the

tool-centre distance sensor still reads $+4.60$ mm (median; s.d. 2.69; $n = 20$ poses), because the rim of the 15 mm-radius disc reaches the curved surface before its centre does. A tolerance written on the tool centre ($\delta = 3$ mm) therefore *excludes* the contact that does the polishing and reports 30–32 % coverage for the same policies whose physically-gated area is 87–88 %—an understatement by a factor of 2.8. All figures below use physical contact.

Table II reports the result at the nominal placement. *Area* is the fraction of the working region's surface swept by the disc while the disc is in contact, *Path completion* is the fraction of the 566 waypoints reached *in order* and *Contact* is the fraction of control steps spent touching the part. Path completion is not reported for the classical pipeline, which runs open loop and never returns to a waypoint it misses, so its in-order completion stalls at 2.1 % while it keeps polishing the rest of the part. Across five seeds, the learned policies polish 88.3 ± 11.2 % of the working region without a safety term and 86.6 ± 9.0 % with the joint-velocity effort term, a difference that is not significant ($p = 0.92$). The distribution of outcomes shows two distinct peaks: two seeds per policy complete the path and reach the ceiling (97.2–97.5 % of the area), while the other three leave the plan early (path completion 0.2–12.7 %) yet still polish 69–96 % of the area, which explains the large deviations of path completion. The classical pipeline reaches 96.1 %. Furthermore, the PPO $+ r_{safe}$ policy sustains a higher contact rate (96.0 %) compared to both the classical pipeline (92.2 %) and the policy with no safety term (76.7 %). This improvement over the policy with no safety term is the only statistically significant difference at this placement ($p = 0.008$).

TABLE II
POLISHED AREA AT THE NOMINAL PLACEMENT

Policy	Area	Path compl.	Contact
Path ceiling	97.6 %	—	—
PPO, no safety term	88.3 ± 11.2 %	41.7 ± 47.7 %	76.7 ± 19.3 %
PPO $+ r_{safe}$	86.6 ± 9.0 %	44.7 ± 45.4 %	96.0 ± 2.0 %
Classical CPP+IK	96.1 %	n/a	92.2 %

B. Kinematic Safety

The task must be kinematically stressed. Fig. 3 carries the evidence. At the nominal placement the path never enters the ill-conditioned region as the damped inverse loses at most 8 % of the commanded motion, and the joint-velocity demand $\max_i |(\mathbf{J}^+ \hat{\mathbf{t}})_i|$, the speed of the fastest joint required per unit of tool speed along the path, never exceeds $3.0\,\mathrm{rad\,m^{-1}}$. Since the joints allow $\dot{q}_{max} = 0.5\,\mathrm{rad\,s^{-1}}$ and the tool is commanded at up to $v_{max} = 0.05\,\mathrm{m\,s^{-1}}$, any demand up to $\dot{q}_{max}/v_{max} = 10\,\mathrm{rad\,m^{-1}}$ is traversable at full speed: the nominal path uses at most 30 % of the joint-velocity capacity, and no safety term can be shown to help.

Moving the part 0.15 m closer to the base leaves the toolpath geometrically unchanged but drives 18.2 % of it below 3λ from the reset posture, with losses of up to 94 % and joint-velocity demands of up to $53\,\mathrm{rad\,m^{-1}}$: 0.4 % of the waypoints demand more than twice what the actuator can deliver at nominal speed. The stress, however, belongs to the posture as much as to the placement: the self-motion only spins the tool and cannot change σ_{min}, but the reset posture with its wrist flipped tracks the whole stressed path above 4λ, below $3.6\,\mathrm{rad\,m^{-1}}$ and free of collision (Fig. 3b). Switching between distinct kinematic solutions during execution requires crossing a singular configuration. The two postures differ in the sign of the wrist angle q_5, so any continuous transition between them passes through $q_5 = 0$, where the axes of the fourth and sixth joints align and σ_{min} vanishes. The region therefore cannot be avoided by a policy starting from the reset posture, even though it could be completely avoided at the planning stage by choosing a different starting posture. The ablation is therefore conducted at the stressed placement, from the reset posture.

The effort term keeps the policy on the plan. Table III reports the ablation at the stressed placement. With r_{safe}, the policy is the only one that continues to follow the prescribed path (63.3% of waypoints, against $\approx$ 1 % for the others, $p \leq 0.010$; three of five seeds complete it, and the other two stall after 6 and 10 % of it) and the only one whose polished area approaches the ceiling (95.3 ± 3.9 %, against 81.6 ± 7.0 % with no safety term and 77.1 ± 3.9 % with the conditioning penalty; $p = 0.032$ and 0.008). The arms without it drift into unstructured sweeping, that is, they still polish around 77–82 % of the area, but abandon the plan. The classical pipeline attains 96.2 % here as well, so the placement remains coverable and the stress adds a learning difficulty. As the classical pipeline matches or exceeds every learned policy in area at both placements, our contribution is the comparison between reward formulations, not an advantage of learning over classical control.

The safety statistics inside the region point the same way, showing the lowest joint velocity (0.223 against 0.261 and $0.282\,\mathrm{rad\,s^{-1}}$), the lowest saturation (10.9 % against 14.6 % and 28.1 %), and the largest elbow margin. However, none of these differences are significant ($p \geq 0.095$). Furthermore, because the arms spend their time in different parts of the workspace, the only demonstrable effect of the effort term is keeping the policy on the plan.

The conditioning-proximity penalty, by contrast, is indistinguishable from carrying no safety term at all on every metric ($p \geq 0.31$), and its saturation inside the region is nominally the highest of the three arms. Given the reset posture, the conditioning along the prescribed path is fixed: the self-motion cannot change it and the waypoints inside the region must be visited, so the penalty can be reduced only by leaving the plan. This conclusion holds only for prescribed coverage from a given posture. It does not extend to conditioning objectives in general, nor to the choice of the initial posture, which removes the region altogether (Fig. 3b).

Fig. 4 shows the behaviour of each case. Without a safety term, the policy cycles repeatedly into the velocity limit for 14.6 % of the steps it spends inside the ill-conditioned regions, as seen in Table III. With the conditioning penalty, it holds for 28.1 %. With the effort term (10.9 %), the highlighted run

saturates once, for about 14 s, on entering the region, then stays below the limit until it completes the path at about 120 s.

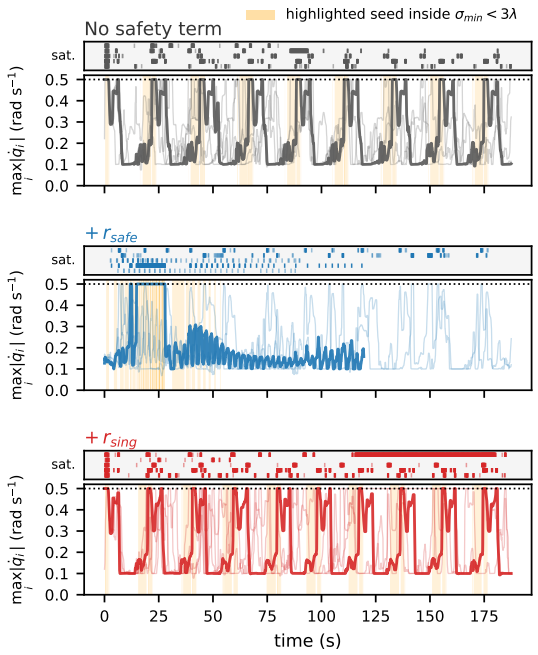


Fig. 4. Commanded joint velocity over the episode at the stressed placement, shown as a rolling 1-s median for the five seeds of each arm. The highlighted seed is the one with the median episode length, and orange shading marks the intervals it spends inside the ill-conditioned region. The strip above each panel marks, for each seed, the steps at the velocity limit, and the dotted line shows that limit, $\dot{q}_{max}$.

TABLE III
SAFETY AT THE STRESSED PLACEMENT

Arm	**Area**	**Path compl.**[c]	**Sat. in**[a]	**Elbow**[b]
No safety term	$81.6 \pm 7.0\,\%$	$1.1 \pm 1.4\,\%$	$14.6 \pm 10.8\,\%$	0.173
$+ r_{safe}$	$95.3 \pm 3.9\,\%$	$\mathbf{63.3 \pm 45.0}\,\%$	$10.9 \pm 12.0\,\%$	0.203
$+ r_{sing}$	$77.1 \pm 3.9\,\%$	$0.9 \pm 1.5\,\%$	$28.1 \pm 19.4\,\%$	0.182
Classical CPP+IK	$96.2\,\%$	n/a	—	0.195

[a] Steps with a joint at its velocity limit, inside $\sigma_{min} < 3\lambda$.
[b] Min. elbow distance to its $\pm\pi$ limit (half the others' travel).
[c] Bold: significant against both other learned policies ($p \leq 0.010$).

VI. CONCLUSION

We presented a reinforcement learning framework for robotic polishing in which the coverage path is generated in the intrinsic metric of the workpiece surface and executed under contact by a 5-DOF operational-space policy. At the nominal placement, the learned policies polish 87–88 % of the working region on average and up to 97.5 % when they complete the path, close to the 97.6 % that perfect tracking would achieve. The classical CPP+IK pipeline reaches 96.1 %, but only with a single distance along the surface normal, chosen through repeated offline runs, whereas the policy with the effort term adjusts contact online and keeps it for a larger fraction of the time (96.0 against 92.2 %). Across seeds, however, training either completes the path or abandons it early, and the classical pipeline matches or exceeds the learned policies in polished area. Under kinematic stress, the joint-velocity effort term is the only reward variant that keeps the policy on the plan, reaching 63.3 % of the waypoints and 95.3 % of the area. A conditioning penalty has no measurable effect, because along a prescribed path the conditioning is determined by the placement and the initial posture, which the task redundancy cannot change.

This study is limited to simulation, a single workpiece, two placements and geometric contact, so high coverage does not yet guarantee polishing quality. The analysis of alternative arm postures is also purely kinematic. Future work will therefore address force-aware control, as in safe variable-impedance learning for polishing [14], and treat the choice of the initial posture as part of planning, moving the framework from geometric coverage toward process quality.

REFERENCES

[1] M. Dalle Vedove, F. J. Abu-Dakka, L. Palopoli, D. Fontanelli, and M. Saveriano, "MeshDMP: motion planning on discrete manifolds using dynamic movement primitives," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2025, pp. 895–901, doi: 10.1109/ICRA55743.2025.11128556.

[2] L. Gong, L. Li, L. Zhang, C. Fan, and N. Li, "Uniform coverage trajectory planning for polishing of equation-free surfaces," *Machines*, vol. 13, no. 7, Art. no. 568, 2025, doi: 10.3390/machines13070568.

[3] V.-T. Do and Q.-C. Pham, "Geometry-aware coverage path planning for depowdering on complex 3D surfaces," *IEEE Robot. Autom. Lett.*, vol. 8, no. 9, pp. 5552–5559, 2023, doi: 10.1109/LRA.2023.3296943.

[4] S. McGovern and J. Xiao, "A general approach for constrained robotic coverage path planning on 3D freeform surfaces," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 4, pp. 5546–5557, 2024, doi: 10.1109/TASE.2023.3313228.

[5] S.-K. Chen and J.-H. Wu, "Intelligent singularity avoidance in UR10 robotic arm path planning using hybrid fuzzy logic and reinforcement learning," in *Proceedings of TANET 2025*, Paper No. T0404, 2025. doi: 10.48550/arXiv.2601.05836.

[6] S. A. Alizadeh Kolagar, M. H. Shahna, and J. Mattila, "Combining deep reinforcement learning with a jerk-bounded trajectory generator for kinematically constrained motion planning," in *Proc. Eur. Control Conf. (ECC)*, 2025, pp. 2507–2514, doi: 10.23919/ECC65951.2025.11187262.

[7] P. Liu, H. Bou-Ammar, J. Peters, and D. Tateo, "Safe reinforcement learning on the constraint manifold: Theory and applications," *IEEE Trans. Robot.*, vol. 41, pp. 3442–3461, 2025, doi: 10.1109/TRO.2025.3567477.

[8] Y. Shen *et al.*, "Reinforcement learning-based reactive obstacle avoidance method for redundant manipulators," *Entropy*, vol. 24, no. 2, Art. no. 279, 2022, doi: 10.3390/e24020279.

[9] K. Crane, C. Weischedel, and M. Wardetzky, "The heat method for distance computation," *Commun. ACM*, vol. 60, no. 11, pp. 90–99, November 2017, doi: 10.1145/3131280.

[10] Y. Wang and M. Gleicher, "Hierarchically accelerated coverage path planning for redundant manipulators," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2025, pp. 12098–12104, doi: 10.1109/ICRA55743.2025.11128545.

[11] C. W. Wampler, "Manipulator inverse kinematic solutions based on vector formulations and damped least-squares methods," *IEEE Trans. Syst., Man, Cybern.*, vol. 16, no. 1, pp. 93–101, 1986, doi: 10.1109/TSMC.1986.289285.

[12] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[13] A. Raffin *et al.*, "Stable-Baselines3: reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021. [Online]. Available: http://jmlr.org/papers/v22/20-1364.html

[14] E. Cramer, L. Jäschke, and S. Trimpe, "CHEQ-ing the Box: Safe Variable Impedance Learning for Robotic Polishing," *IFAC-PapersOnLine*, vol. 59, no. 18, pp. 325–330, 2025, doi: 10.1016/j.ifacol.2025.10.241.